

EEE5544 Stochastic Methods for Engineering 1

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0.1 Introduction

0.1.1 Introduction

What is *random*?

- Some behavior that we cannot perfectly predict, as oppose to being *deterministic*.

Why random?

- **Theologian:** We can't know God's mind!
- **Copenhagen quantum physicist:** The laws of quantum physics are fundamentally random.

What to do with random?

Although we cannot perfectly determine a random behavior, we still want to be able to obtain to some degree its specifications:

- **Probability Theory** is the theory that systematically studies and characterizes randomness.
- We will go through an engineering (more operational, less about math vigor) treatment of Probability Theory in this course.

How to study probability?

1. Intuition

- **Example:** Dr. Wong is a nice guy, he will probably give me a good grade in EEE 5544 .
- Not systematic, guessing, cannot quantify.

2. Relative frequency (a.k.a. counting)

- **Example:** Dr. Wong taught EEE 5544 three times in the past. There were altogether 200 students who took Dr. Wong's classes. Out of these 200 students, 50 of them got the *A* grade. Thus the chance of getting an *A* in Dr. Wong's class is $\frac{50}{200} = \frac{1}{4}$.
- Assume every grade is "equally likely", which may not be the case.
- What about if there are infinite possibilities?

3. Axiomatic theory

- Developed by Kolmogorov in the early 1900s.
- Identify a few basic axioms that fit intuition.
- Build a math consistent system based on the axioms.
- Utilize measure theory to construct the math system.
- This is the approach that we will follow.

0.2 Probability Space

0.2.1 Probability Space

Random Experiment

- A **random experiment** is an experiment in which the outcome varies in an unpredictable fashion when the experiment is repeated under the same conditions, i.e., *random*.
- An **outcome** of a random experiment is a result that cannot be decomposed into other results.

Sample Space

- The set of all possible outcomes for a random experiment is called the **sample space** Ω .

Examples:

4. Tossing a coin: $\Omega = \{H, T\}$
 5. Tossing a coin twice: $\Omega = \{HH, HT, TH, TT\}$
 6. Taking EEE 5544 and getting a grade: $\Omega = \{A, A-, B+, B, B-, C+, C, D, F\}$
 7. Picking an integer at random: $\Omega = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
 8. Checking the temperature (in K) at a random location $\Omega = (0, \infty)$
- **Finite** Ω : Examples 1, 2, 3
 - **Countably infinite** Ω : Example 4
 - **Discrete** Ω : Examples 1–4
 - **Continuous (uncountably infinite)** Ω : Example 5

Event Class

- An **event** is a set of outcomes (a subset of Ω) that we can “quantify its likelihood” (see below).
- Two special events:
 - $\Omega = \text{certain event}$
 - $\emptyset = \text{impossible event}$
- An **event class** $\mathcal{F}$ of a random experiment is a collection of events.

Discrete sample space

- For a discrete Ω , we typically consider the event class that is the collection of all subsets of Ω , i.e., $\mathcal{F} = 2^\Omega$, which is called the *power set* of Ω .

Examples:

9. Tossing a coin: $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \{HT\}\}$.
10. Tossing a coin twice:

$$\mathcal{F} = \{\emptyset, \{HH\}, \{TT\}, \{HT\}, \{TH\}, \{HH, TT\}, \{HH, HT\}, \dots, \{HH, TT, HT\}, \dots, \{HH, TT, HT, TH\}\}.$$

11. If $\Omega = \{a_1, a_2, \dots, a_N\}$, then

$$\mathcal{F} = \{\emptyset, \{a_1\}, \dots, \{a_N\}, \{a_1, a_2\}, \dots, \{a_1, a_2, a_3\}, \dots, \{a_1, a_2, \dots, a_N\}\}.$$

Altogether 2^N events (why?).

12. For a countably infinite $\Omega = \{a_1, a_2, \dots\}$, we can use the same idea as in the finite case using the power set as the event space, i.e.,

$$\mathcal{F} = 2^\Omega = \{\emptyset, \{a_1\}, \dots, \{a_N\}, \{a_1, a_2\}, \dots, \{a_1, a_2, a_3\}, \dots, \{a_1, a_2, \dots, a_n\}, \dots\}$$

by enumerating all subsets of Ω . However, we need a better way to describe the power set.

Continuous sample space

- For a continuous sample space, e.g., $\Omega = (0, \infty)$, there are uncountably infinite number of outcomes and hence the method of enumeration used in the finite case can't be used to construct the event class anymore.
- Luckily mathematicians have solved the problem for us. To discuss the solution, we need to study a bit about set operations.

Set operations

- **Union:** $A \cup B = \{x \in \Omega : x \in A \text{ or } x \in B\}$
- **Intersection:** $A \cap B = \{x \in \Omega : x \in A \text{ and } x \in B\}$
- **Complement:** $A^c = \{x \in \Omega : x \notin A\}$
- **Difference:** $A \setminus B = \{x \in \Omega : x \in A \text{ and } x \notin B\} = A \cap B^c$

Proof: We give the proof of 1. as an example here:

(i) Since $x \in (A \cup B)^c \Rightarrow x \notin A \cup B \Rightarrow x \notin A \text{ and } x \notin B \Rightarrow x \in A^c \text{ and } x \in B^c \Rightarrow x \in A^c \cap B^c$, $(A \cup B)^c \subseteq A^c \cap B^c$.

(ii) Since $x \in A^c \cap B^c \Rightarrow x \in A^c \text{ and } x \in B^c \Rightarrow x \notin A \text{ and } x \notin B \Rightarrow x \notin A \cup B \Rightarrow x \in (A \cup B)^c$, $A^c \cap B^c \subseteq (A \cup B)^c$.

Combining (i) and (ii), we have $(A \cup B)^c = A^c \cap B^c$.

Field (Algebra)

- An **field (algebra)** is a collection of subsets of Ω that is closed under union and complement, and contains both $\emptyset$ and Ω . That is, $\mathcal{F}$ is a field (algebra) if it satisfies the following three conditions:
 - $\emptyset \in \mathcal{F}$ and $\Omega \in \mathcal{F}$.
 - If $A \in \mathcal{F}$ and $B \in \mathcal{F}$, then $A \cup B \in \mathcal{F}$.
 - If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
- Operationally speaking, a field contains all subsets of Ω that can be obtained from applying a finite number of set operations.

σ -Field (σ -Algebra)

- A field $\mathcal{F}$ is a σ -field if it satisfies the additional condition:
 - If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.
- Operationally speaking, a σ -algebra contains all subsets of Ω that can be obtained from a countably number of set operations.

Generating σ -field

- For a continuous (uncountably infinite) sample space Ω , we often *generate* σ -fields from some collections of subsets of Ω .
- For a collection $\mathcal{A}$ of subsets of Ω , consider the σ -field generated by $\mathcal{A}$, i.e., the smallest σ -field containing all subsets of Ω that can be formed by performing countable set operations on subsets in $\mathcal{A}$.
- The σ -field generated by $\mathcal{A}$ is denoted by $\sigma(\mathcal{A})$.

Event Class for Continuous Sample Space

- For a continuous (uncountably infinite) sample space Ω , we will consider event classes that are *sigma*-fields of Ω .

Example: $\Omega = \mathbb{R}$

- Consider the collection $\mathcal{A}$ of all the subsets of $\mathbb{R}$ in the form of $(-\infty, a)$ for some $a \in \mathbb{R}$.
- The σ -field generated by $\mathcal{A}$, called the *Borel σ -field*, is commonly used as an event class for the sample space $\mathbb{R}$.
- The Borel σ -field contains essentially all events of practical interest.
- While the *Borel sets* (i.e., subsets in the Borel σ -field) form a very rich class of subsets of $\mathbb{R}$, there are subsets of $\mathbb{R}$ that are not Borel sets. This turns out to be critically important in allowing the “likelihood” of the events to be “quantifiable” as we discussed before.

Description of Random Experiment

- We can now use the sample space Ω and an event class $\mathcal{F}$ to mathematically describe a random experiment.
- In addition, we need a way to “quantify the likelihood of the events” as alluded to above. This is done by a mapping called *probability measure* P , which will be discussed in the next section.
- In summary, the math object that describes a random experiment is the triple $(\Omega, \mathcal{F}, P)$ called the **probability space** of the random experiment.